REPURPOSING COLLABORATION OPPORTUNITY

ANAGRELIDE -> type 2 diabetes mellitus

For clinical-collaborator outreach.

Why this hypothesis is real

Drug-target convergence on the disease scores **0.90** (noisy-OR over OT target-disease associations). This pair is fully novel relative to all known drug-indication assignments in ChEMBL. Lead target is well expressed in the disease-relevant tissue (thyroid gland). Orthologous-gene model-organism evidence is strong (phylo_score 0.85, n=9). Reactome co-membership reinforces the indirect mechanism with **1** shared specific pathways.

Why now (IP runway + literature footprint)

The substance is a biologic or otherwise outside the FDA Orange Book; check the Purple Book and direct registry for IP runway.

Clinical opportunity

US population: approximately **37,300,000** patients -- a mass-market opportunity. (source: CDC National Diabetes Statistics Report)

What we propose

BioBix (Department of Mathematical Modelling, Statistics and Bioinformatics, Ghent University) brings bioinformatics support to the collaboration:
• Target validation across blood / immune / cancer multi-omics datasets
• Biomarker candidate identification from public + private cohorts
• Patient stratification analytics for trial-enrichment design
• Mechanism-of-action mining via systems-biology priors
• Quick-turn evidence syntheses for IIT and partnered programs

Proposed collaboration model:

• Investigator-initiated trial framework with the substance owner
• BioBix as bioinformatics analytics partner (target/biomarker/biostatistics)
• Joint publication strategy; data-sharing agreement gated on IP terms

Next step

A 30-minute scoping call to align on feasibility. We would come prepared with a brief mechanistic dossier, an annotated literature pull, and a pilot biomarker shortlist for the disease.

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